

DEVAYAN MANDAL

Full Stack AI Native / Forward Deployed AI Engineer & Open Source Contributor - Open to Remote Work

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PROFESSIONAL SUMMARY

Full Stack AI Native / Forward Deployed AI Engineer with 10 years of remote, self-directed production work. Built 30+ production-grade systems across autonomous multi-agent platforms, MCP integrations, deterministic RL evaluation, real-time observability, and high-throughput web applications. Merged 21 upstream pull requests across TypeScript, JavaScript, Python, and Rust in Recharts, pnpm, node-postgres, SQLMesh, Biome, Better Auth, thirdweb, Anza Kit, and Rspress. Specializes in explicit state machines, bounded autonomy, verifiable AI systems, hybrid RAG and memory architectures, streaming infrastructure, and production-grade developer tooling, with end-to-end ownership from architecture through deployment, observability, testing, and upstream delivery.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Rust, SQL, R, Bash

AI & Agents: LangGraph, LangChain, LangServe, LangSmith, Model Context Protocol (MCP), multi-agent orchestration, supervisor routing, Reflexion loops, human-in-the-loop gates, hybrid RAG, RLVR environments, deterministic evaluation, prompt engineering, Ollama, Gemini, OpenAI-compatible providers

Frontend: React 19, Next.js 16, TanStack Start/Router/Query, Tailwind v4, Vite, Framer Motion, Expo, Three.js, React Three Fiber, HTML5 Canvas, SVG

Backend & Data: FastAPI, Node.js, Drizzle ORM, Prisma, SQLAlchemy, PostgreSQL (pgvector), Neon, Supabase, SQLite, Pydantic, Zod, Stripe

Realtime & Observability: SSE, WebSockets, Socket.io, replayable traces,

OpenTelemetry-shaped tracing, LangSmith evaluation pipelines

Cloud & DevOps: Vercel, AWS, Microsoft Azure, GitHub Actions CI/CD, serverless, scheduled pipelines, Vitest, pytest, Playwright, Biome

Web3 & Blockchain: Solidity, OpenZeppelin, Foundry, viem, wagmi, ethers.js, Safe Protocol Kit, RainbowKit, Reown AppKit, Base L2, blockchain and wallet infrastructure

Practices: Six Sigma DMAIC, bounded autonomy, retrieval-bounded generation, AST-sandboxed grading, cost-aware routing, threat-model documentation, Git, Changesets

PROFESSIONAL EXPERIENCE

LATTICE AI LABS

Full Stack AI Native / Forward Deployed AI Engineer | January 2016 - Present | Remote

- Architected and shipped 30+ production-grade systems across a broad portfolio of public engineering repositories spanning autonomous agents, RL evaluation environments, real-time observability, and high-throughput web applications.

- Built AEGIS, an enterprise AI ops cortex where a LangGraph supervisor routes 6 specialist agents across 7 agentic loops, with hybrid RAG, human-in-the-loop gates, LangSmith evaluation gates, and a live SSE dashboard. Live: aegis-agent-api.vercel.app/ui

- Built Synthesis, an autonomous research platform running cyclic Reflexion workflows (Plan, Research, Synthesize, Critique, Finalize) on a StateGraph executor with HITL approval gates, and AAROP, an inspectable 6-phase reasoning engine with cost-aware model routing, circuit breakers, RAG memory, replayable traces, and 99% Python test coverage. Live: synthesis-gold.vercel.app | aarop.vercel.app

- Built Veritas, a LangGraph research agent with SSE streaming, grounded demo mode, external retrieval when configured, and explicit separation between deterministic demo execution and live model/retrieval execution. Live: veritas-engine-woad.vercel.app

- Authored 3 official RLVR environments on the Prime Intellect Environments Hub (calendar-math, meeting-slot, unit-fix) using deterministic UTC sweep-line solvers and AST-sandboxed execution graders, removing LLM-as-judge variance.

- Implemented MCP servers and clients across Stdio and StreamableHTTP/SSE transports with client-side sampling and roots-based security boundaries; shipped HoldSlot, a WebMCP implementation for the OpenAI WebMCP Challenge 2026.

- Engineered Cadence, a framework-free healthcare agent with 5-layer clinical memory (session, episodic, semantic, knowledge, insights) using Ollama and SQLite, producing consent-scoped clinician briefs. Live: cadence-healthcare.vercel.app

- Shipped Jobrow, a live US tech job board indexing 5,000+ still-open roles across employer ATS boards (Greenhouse, Ashby, Lever) on TanStack Start, React 19, and Neon Postgres, with a public JSON API, Expo iOS/Android client, and GitHub Actions crawl pipeline running 4x daily. Live: jobrow.vercel.app
- Shipped Lattice, a live Web3 job platform for blockchain, crypto, and DeFi roles with same-day close detection from employer ATS boards. Live: lattice-devtechedge1.vercel.app
- Built a real-time chaos engineering platform with self-healing microservices, animated SVG topology, automated fault injection, and sub-second WebSocket telemetry, plus CollabSpace multiplayer whiteboards on Supabase Realtime; wired replayable OTel-shaped traces and LangSmith evaluation pipelines through the agent systems. Live: chaos-simulation.vercel.app
- Delivered Pulsar, a Base L2 GPU marketplace with Foundry smart contracts; RegTrace-AI, a MiCA/VARA compliance copilot with source-traced citations and hallucination flags; Smart Waitlist with Stripe-backed referral flows; and AstraMart.
- Operate a serverless daily AI news agent on GitHub Actions that pulls six public feeds, dedupes against in-repo memory, summarizes with Gemini, and delivers one Telegram executive brief at zero infrastructure cost.
- Work AI-native across the full stack using frontier coding and reasoning models through dedicated development harnesses, with emphasis on rapid iteration, test-driven validation, repository-specific contribution policies, and reproducible upstream delivery.

SPECTRAL QUANTITATIVE TRADING

Quantitative Crypto Trader & Derivatives Strategist | September 2016 - Present | Remote

- Manage diversified proprietary capital across tier-1 exchanges including Binance, Coinbase, Bitget, and MEXC, and on-chain protocols including Hyperliquid, Uniswap v3, Aave, and GMX.
- Built and ran automated grid, DCA, and Martingale bots on Pionex and Binance to capture spread and volatility in ranging markets.
- Executed delta-neutral basis and funding-rate arbitrage between spot and perpetuals, plus concentrated-liquidity and lending-loop strategies on-chain.
- Applied TWAP/VWAP execution on illiquid pairs, with VaR limits, drawdown caps, and Sharpe-ratio optimization governing position sizing and portfolio risk.

OPEN SOURCE CONTRIBUTIONS

Merged upstream:

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- leo-aa88/reviewgate #144 (TypeScript) - adds a configurable excessive code-comment verbosity check to the core review engine: oversized_comment_block, excessive_comment_lines and comment_heavy_diff are computed from added patch lines only, so a PR is never penalised for comments it did not touch. A single-pass lexer tracks string literals, block comments, JS/Go template literals, shell heredocs and Python triple-quoted strings, so runtime data such as a banner inside a backtick string is never miscounted as commentary, and files in languages with no scanner profile contribute to the ratio denominator only. Hunk entry state is established from diff context that carries a code token, which keeps a verbose block added mid-file in an existing file detectable while docstring content stays excluded, and all three dimensions feed the existing baseline_reviewability aggregation through the policy.code_comments config block. Merged Sep 2026. github.com/leo-aa88/reviewgate/pull/144
- PyO3/maturin #3302 (Rust) - Windows interpreter discovery no longer panics on non-UTF-8 command output. find_all_windows() decodes py --list-paths and conda info -e with String::from_utf8_lossy instead of unwrapping str::from_utf8, so a localized install that emits undecodable bytes skips that line instead of aborting the build. Added a unit test covering the non-UTF-8 decode path. Merged Sep 2026. github.com/PyO3/maturin/pull/3302
- stellar/stellar-docs #2859 (MDX) - added the --enable-core-manual-close flag to the Local section of the Network Modes page, so the parameter list no longer omits a flag the quickstart container ships, with a cross-link to the Manual close mode section noting that only the local configuration sets NODE_IS_VALIDATOR. Merged Sep 2026. github.com/stellar/stellar-docs/pull/2859
- stellar/stellar-docs #2850 (MDX) - documented Quickstart's undocumented '--enable-core-manual-close' flag in the advanced usage docs, covering the 'MANUAL_CLOSE' setting it writes into the generated core config, the 'manualclose' endpoint that advances one ledger per call, and loopback-bound startup examples for macOS, Linux, and Windows. Merged Sep 2026. github.com/stellar/stellar-docs/pull/2850
- ssf0409/tracelens #140 (Python) - markdown table cells in the report generator are now escaped with html.escape in addition to pipe and newline handling. Task ids or gate values

containing pipes, line breaks, or HTML metacharacters no longer break the per-task and baseline-gate tables piped into \$GITHUB_STEP_SUMMARY, and no cell can open a raw HTML element. Merged Sep 2026. github.com/ssf0409/tracelens/pull/140

- stellar/stellar-docs #2849 (MDX) - reworked the address conversion example to propagate the fallible 'Result<Address, ConversionError>' instead of unwrapping malformed XDR, and documented that untrusted XDR must be handled rather than panicking. Merged Sep 2026. github.com/stellar/stellar-docs/pull/2849
- stellar/stellar-docs #2853 (MDX) - reconciled memo framing in the pooled accounts guide so memos read as the legacy mechanism still in active use and muxed accounts as preferred going forward. Merged Sep 2026. github.com/stellar/stellar-docs/pull/2853
- stellar/stellar-docs #2851 (MDX) - qualified the Freighter HTTPS requirement, noting that 'http://localhost' is already a Potentially Trustworthy origin under W3C Secure Contexts so loopback needs no TLS. Merged Sep 2026. github.com/stellar/stellar-docs/pull/2851
- pytest-dev/pytest-env #262 (Python) - documented that pytest.toml and .pytest.toml accept the native [pytest] env table, not only the plugin-specific [pytest_env] section. Merged Sep 2026. github.com/pytest-dev/pytest-env/pull/262
- pnpm/pnpm #14863 (Rust/TypeScript) - fixed startup crashes on FreeBSD and other non-Windows Unix-like platforms by making default_store_dir use the Unix fallback path; added platform-specific regression coverage. Merged Sep 2026. github.com/pnpm/pnpm/pull/14863
- web-infra-dev/rspress #3678 (TypeScript) - fixed a Search initialization race by awaiting FlexSearch addAsync indexing before LocalProvider.init() resolves, preventing early searches from incorrectly returning no results on larger sites; focused regression-oriented fix. Merged Sep 2026. github.com/web-infra-dev/rspress/pull/3678
- recharts/recharts #7805 (TypeScript) - removed tabIndex={-1} from z-index portal SVG <g> layers so empty groups are not pointer-focusable and WebKit does not draw geometry-traced focus rings; added regression coverage asserting no layer carries a tabIndex attribute. Merged Sep 2026. github.com/recharts/recharts/pull/7805
- brian/node-postgres #3772 (JavaScript) - stopped Connection.sync() from incorrectly setting the internal _ending flag, preventing false suppression of subsequent socket errors; added unit coverage for Sync vs disconnect semantics. Merged Sep 2026. github.com/brian/node-postgres/pull/3772
- pnpm/pnpm #14756 (Rust/TypeScript) - preserved existing dependency range operators and npm:/jsr: protocol prefixes during pnpm update, including prerelease and exact-version handling. Merged Sep 2026. github.com/pnpm/pnpm/pull/14756
- pnpm/pnpm #14754 (Rust) - fixed non-recursive pattern runs with --no-bail so matching scripts continue executing and failures are aggregated correctly; added concurrent and sequential regression tests. Merged Sep 2026. github.com/pnpm/pnpm/pull/14754
- pnpm/pnpm #14753 (Rust/TypeScript) - fixed lockfile:false being ignored when automatic package-manager switching downloaded the configured engine; added unit and filesystem-backed regression coverage across Rust and TypeScript paths. Merged Sep 2026. github.com/pnpm/pnpm/pull/14753
- SQLMesh/sqlmesh #6040 (Python) - moved ModelTest.create_test() off concurrent worker threads to eliminate a time_machine/ttl_cache race in sqlmesh test; verified the focused race regression repeatedly. Merged Sep 2026. github.com/SQLMesh/sqlmesh/pull/6040
- anza-xyz/kit #2032 (TypeScript) - restored number predicates in the getPatternMatchCodec advanced guide to match current codec typing. Merged Sep 2026. github.com/anza-xyz/kit/pull/2032
- better-auth/better-auth #11208 (TypeScript) - added regression coverage locking the /phone-number/verify OpenAPI requestBody after a Zod .and(z.record()) intersection; tests only, no production generator changes. Merged Sep 2026. github.com/better-auth/better-auth/pull/11208
- biomejs/biome #11667 (Rust) - ported the unicorn better-dom-traversing rule into Biome's nursery linter; +1,972 lines across 17 files and 7 commits, released to 13 packages via changeset. Merged Sep 2026. github.com/biomejs/biome/pull/11667
- thirdweb-dev/js #8938 (TypeScript) - fixed useTokenQuery error handling so 401/429/timeout failures rethrow instead of rendering a false "Token Not Supported" state in CheckoutWidget; added tests and a patch changeset. Merged Sep 2026. github.com/thirdweb-dev/js/pull/8938

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Active upstream engineering spans AI infrastructure, developer tooling, package management, frameworks, accessibility, concurrency, portability, security, SDKs, wallets, and blockchain infrastructure. Canonical ledger: github.com/devtechedge/oss-contributions

CERTIFICATIONS

- Model Context Protocol: Advanced Topics - Anthropic, 2026
- AI Fluency for Builders - Anthropic (with CodePath), 2026
- Agents and Workflows - OpenAI Academy, 2026
- Deep Agents with LangGraph - LangChain Academy, 2026
- Ambient Agents with LangGraph - LangChain Academy, 2026
- Microsoft Certified: Azure Fundamentals (AZ-900) - Microsoft, 2020
- AWS Fundamentals Specialization - Amazon Web Services (Coursera), 2020
- Data Science Specialization - Johns Hopkins University (Coursera), 2020
- Six Sigma Green Belt - Kennesaw State University (Coursera), 2020

EDUCATION

UNIVERSITY OF MUMBAI

Bachelor of Engineering (B.E.), Chemical Engineering (First Class) | [Graduation Year]

Quantitative core in process dynamics, transport phenomena, mathematical modeling, and optimization; transitioned into software engineering through computational and systems-oriented work.